Equalizer Lowpass FIR Filter Design via Convex Optimisation

1 Overview

A complex-valued lowpass FIR filter is designed whose passband target matches the UCDC equalizer frequency response. The filter is found by solving a second-order cone programme (SOCP) using CVXPY.

Signal chain parameters

Parameter	Value
Sample rate f_s	750 MHz (two-sided, Nyquist = ±375 MHz)
Passband edge f_p	±162.5 MHz
Stopband edge f_{stop}	±250.0 MHz
Filter taps N	63
Passband tolerance δ_p	0.05
Stopband tolerance δ_s	0.1

2 Optimisation Formulation

The DTFT of the complex tap vector $\mathbf{h} \in \mathbf{C}^N$ is

$$H(\omega) = \sum_{n=0}^{N-1} h[n]\, e^{-j\omega n} = \mathbf{a}(\omega)^\top \mathbf{h}, \quad \mathbf{a}(\omega) = \left[1,\, e^{-j\omega},\, \ldots,\, e^{-j\omega(N-1)}\right]^\top.$$

Sampling over a frequency grid turns this into the linear map $\mathbf{H} = \mathbf{A}\mathbf{h}$, where $A_{kn} = e^{-j\omega_k n}$.

Let $\mathcal{P}$, $\mathcal{S}$, and $\mathcal{T}$ denote the index sets of the passband, stopband, and transition-band frequency grid points, respectively. The **fixed-tolerance** problem is

$$\begin{aligned}
\underset{\mathbf{h}}{\text{minimise}} \quad & \sum_{k\in\mathcal{P}} |\mathbf{a}(\omega_k)^\top \mathbf{h} - H_d(\omega_k)| + \lambda\|\mathbf{h}\|^2 \\
\text{subject to} \quad & |\mathbf{a}(\omega_k)^\top \mathbf{h} - H_d(\omega_k)| \le \delta_p, \quad k \in \mathcal{P} \\
& |\mathbf{a}(\omega_k)^\top \mathbf{h}| \le \delta_s, \quad k \in \mathcal{S} \\
& |\mathbf{a}(\omega_k)^\top \mathbf{h}| \le \delta_{t,k}, \quad k \in \mathcal{T}.
\end{aligned}$$

The transition-band bound $\delta_{t,k}$ decreases linearly from $\max_{k\in\mathcal{P}} |H_d(\omega_k)|$ at the passband edge to $3\delta_s$ at the stopband edge, preventing large overshoot in the unconstrained gap while keeping the SOCP feasible.

3 Filter Response

Figure 1 shows the designed filter's magnitude (dB) and phase response overlaid with the desired equalizer response $H_d(\omega)$.

Figure 2 shows the in-band magnitude response zoomed to ±5 MHz beyond the passband edges, with the y-axis limited to $[-0.9, +0.1]$ dB to resolve the passband ripple.

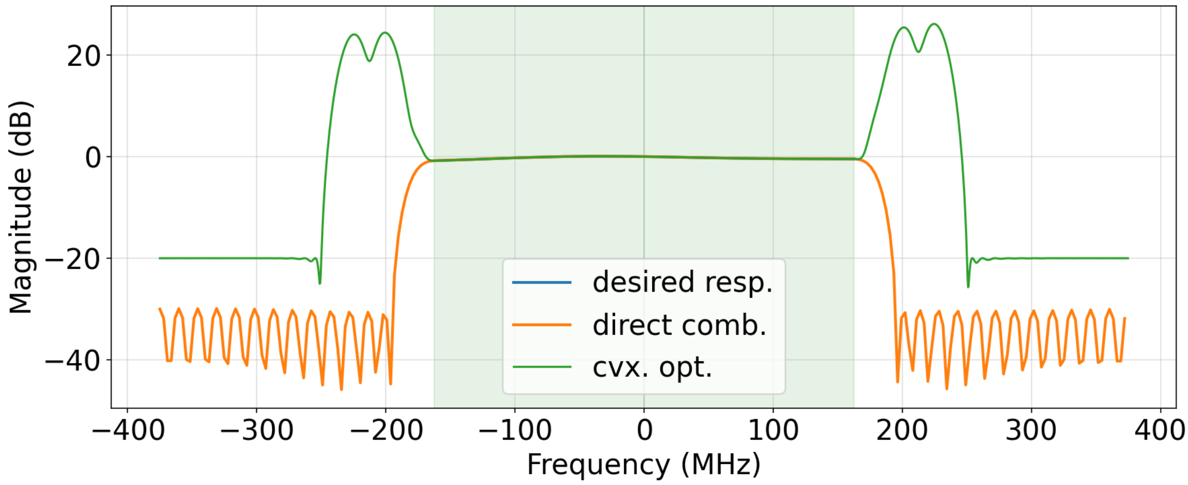

Figure 1: Magnitude (top) and phase (bottom) of the designed equalizer lowpass FIR filter. The green shaded regions indicate the passband ±162.5 MHz. The dashed red line is the desired response $H_d(\omega)$ interpolated from the 256-bin UCDC equalizer FFT.

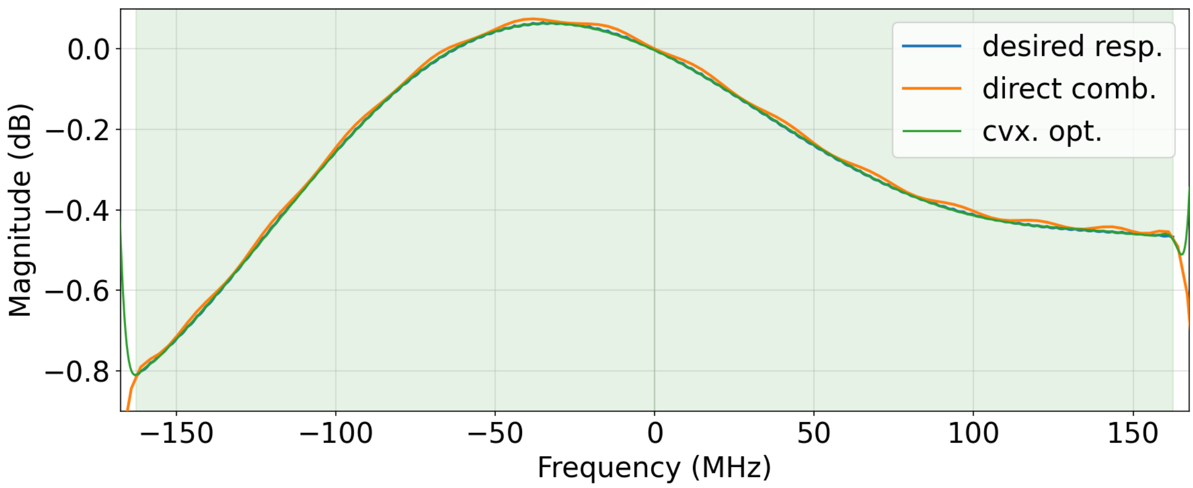

Figure 2: In-band magnitude response zoomed to $[-167.5, +167.5]$ MHz with a $[-0.9, +0.1]$ dB y-axis. The green shaded regions indicate the passband ±162.5 MHz.

4 Complex Remez Algorithm

The complex Remez algorithm (also known as Lawson's iteratively reweighted least-squares, IRLS) extends the classical Parks–McClellan algorithm to complex-valued FIR filters. At each iteration the algorithm solves a weighted least-squares problem

$$\min_{\mathbf{h}} \sum_{k} w_k(\ell) \left| \mathbf{a}(\omega_k)^\top \mathbf{h} - H_d(\omega_k) \right|^2,$$

and then updates the weights as $w_k(\ell+1) = w_k(\ell)\,|e_k(\ell)|$, where $e_k(\ell)$ is the error at frequency ω_k after iteration ℓ. This reweighting progressively concentrates the design effort on the worst-case frequency points, driving the solution toward the minimax (Chebyshev) optimum.

Unlike the convex optimisation approach, no explicit passband or stopband tolerance parameters are required; the algorithm converges to the equi-ripple solution determined solely by the frequency grid and the desired response $H_d(\omega)$.

Figure 3 shows the full-band magnitude and phase response of the filter designed with the complex Remez algorithm. Figure 4 zooms into the passband region, and Figure 5 shows the in-band approximation error.

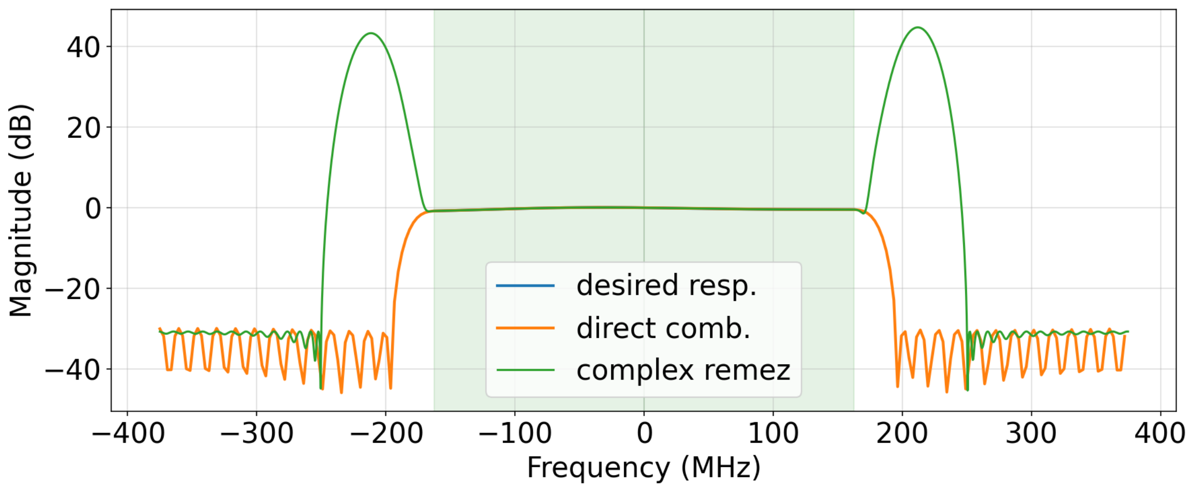

Figure 3: Full-band magnitude (top) and phase (bottom) of the complex Remez FIR filter. The green shaded region indicates the passband ±162.5 MHz. The dashed red line is the desired response $H_d(\omega)$.

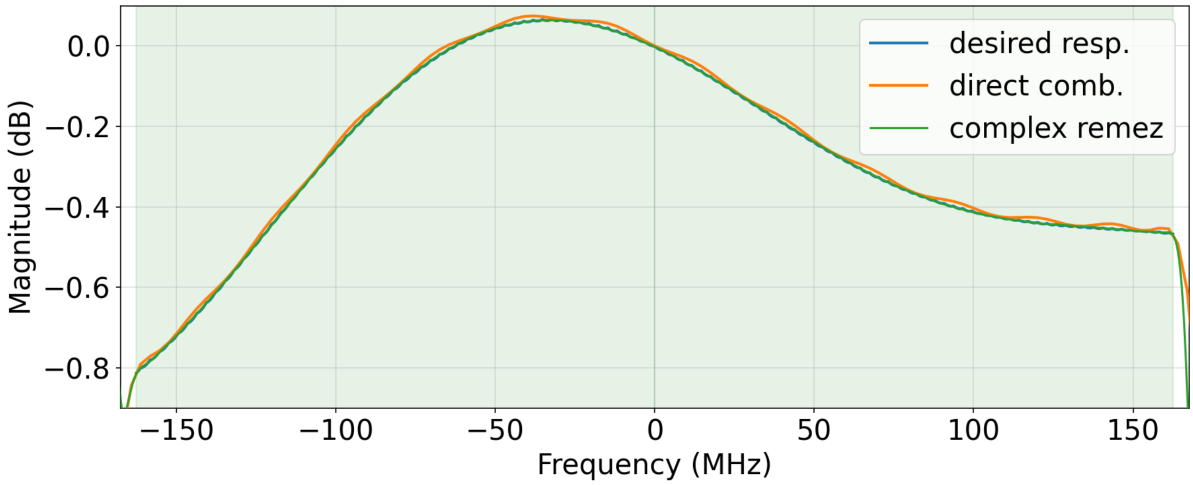

Figure 4: In-band magnitude response of the complex Remez filter zoomed to the passband region. The green shaded region indicates the passband $\pm 162.5\,\mathrm{MHz}$.

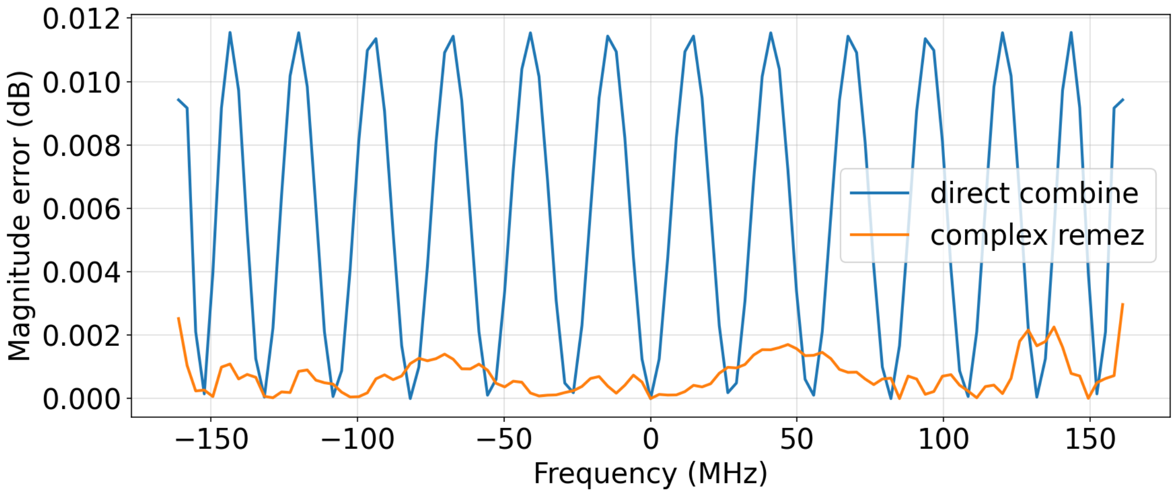

Figure 5: In-band approximation error of the complex Remez filter. The equi-ripple characteristic of the Chebyshev solution is visible across the passband.

4.1 Cosine Transition-Band Constraint

Without a transition-band constraint the IRLS solver leaves the gap between the passband and stopband entirely unconstrained, which can cause the magnitude to overshoot or roll off slowly in that region. A soft constraint is introduced by appending a transition-band grid $\mathcal{T}$ to the unified frequency grid and supplying a cosine-tapered desired magnitude $H_d^{(\text{tb})}(\omega)$ that interpolates smoothly between the passband-edge level and the stopband target. Specifically, for a normalised parameter $t \in [0, 1]$ sweeping from the passband edge to the stopband edge,

$$H_d^{(\text{tb})}(t) = A_{\text{stop}} + \left(A_{\text{pass}} - A_{\text{stop}}\right) \cdot \frac{1 + \cos(\pi t)}{2},$$

where $A_{\text{pass}} = \max_{k \in \mathcal{P}} |H_d(\omega_k)|$ is the passband-edge magnitude and $A_{\text{stop}} = \delta_s$ is the stopband target. This raised-cosine shape has zero derivative at both endpoints, giving a gentle rolloff at the passband edge and a smooth approach to the stopband level without abrupt kinks. The transition-band points enter the IRLS cost with the same band weight as the stopband ($B_k = w_{\text{sb}}$), so the solver tracks the ramp target directly rather than simply suppressing energy toward zero.

Figures 6–8 show the filter designed with this cosine transition-band constraint.

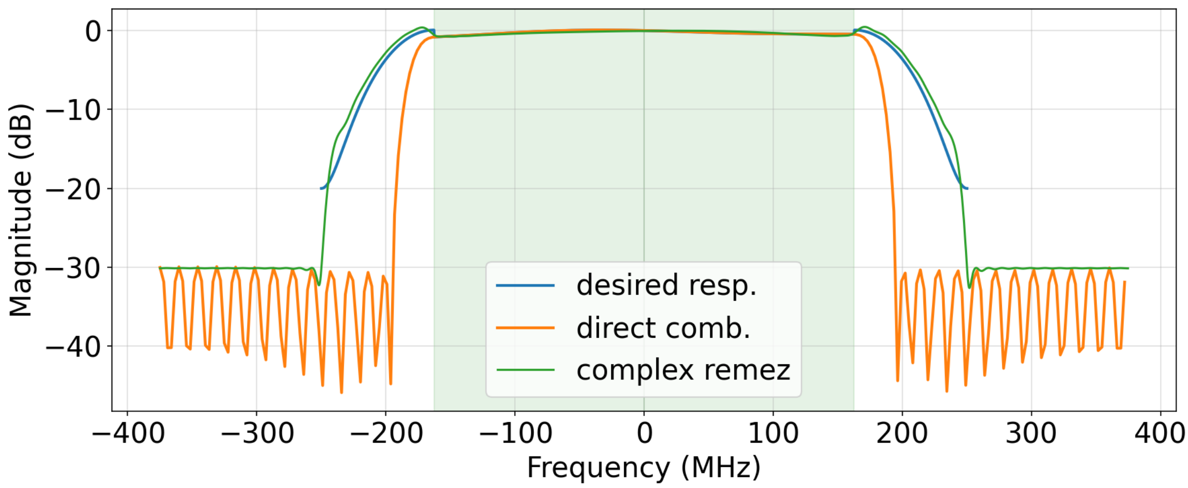

Figure 6: Full-band magnitude and phase of the complex Remez filter with cosine transition-band constraint. Compared with the unconstrained design (Figure 3), the transition region rolls off more smoothly between ±162.5 MHz and ±250 MHz.

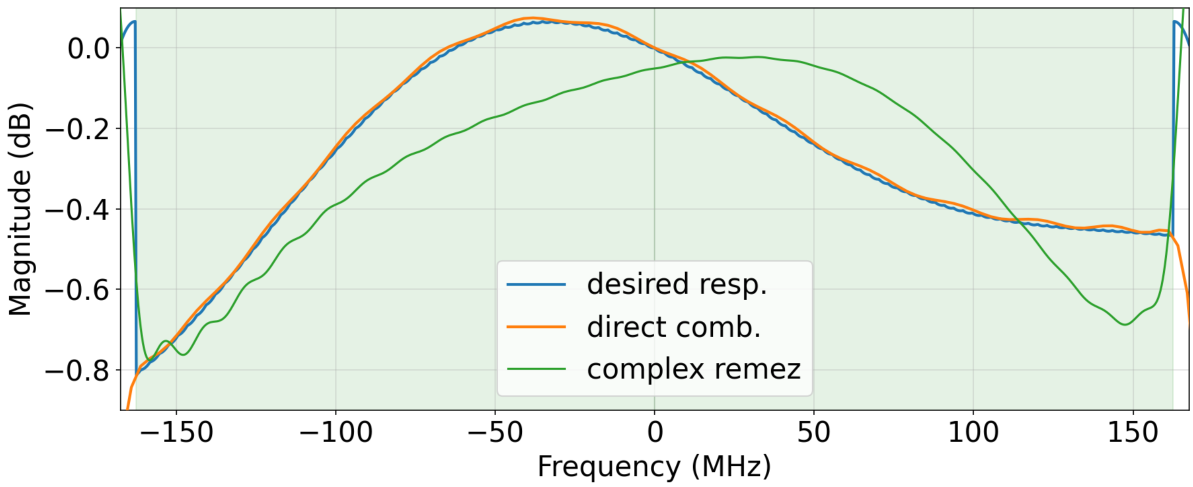

Figure 7: In-band magnitude response of the cosine-constrained Remez filter zoomed to the passband region.

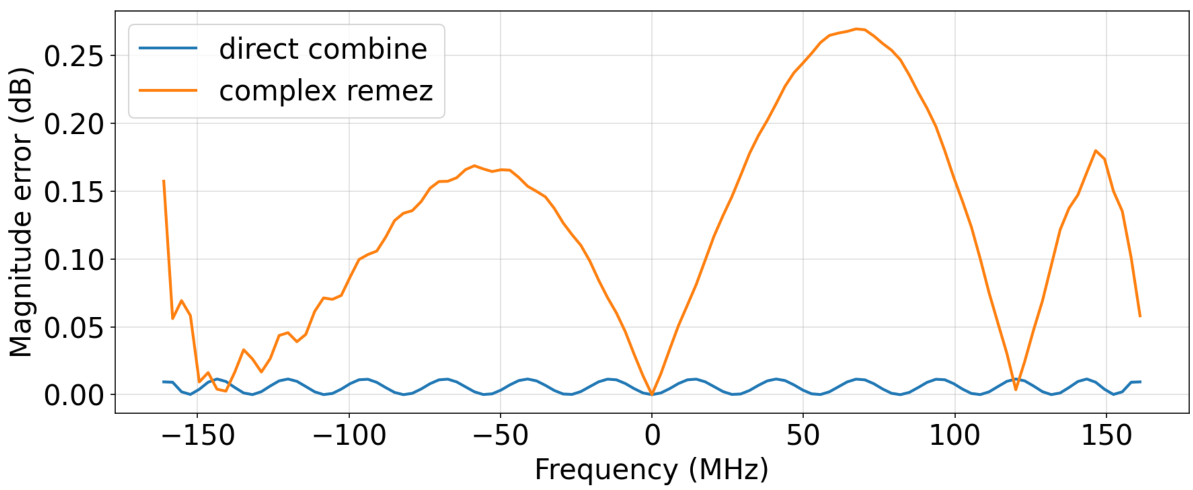

Figure 8: In-band approximation error of the cosine-constrained Remez filter. The additional transition-band degrees of freedom redistribute the equi-ripple error across the passband relative to the unconstrained result.